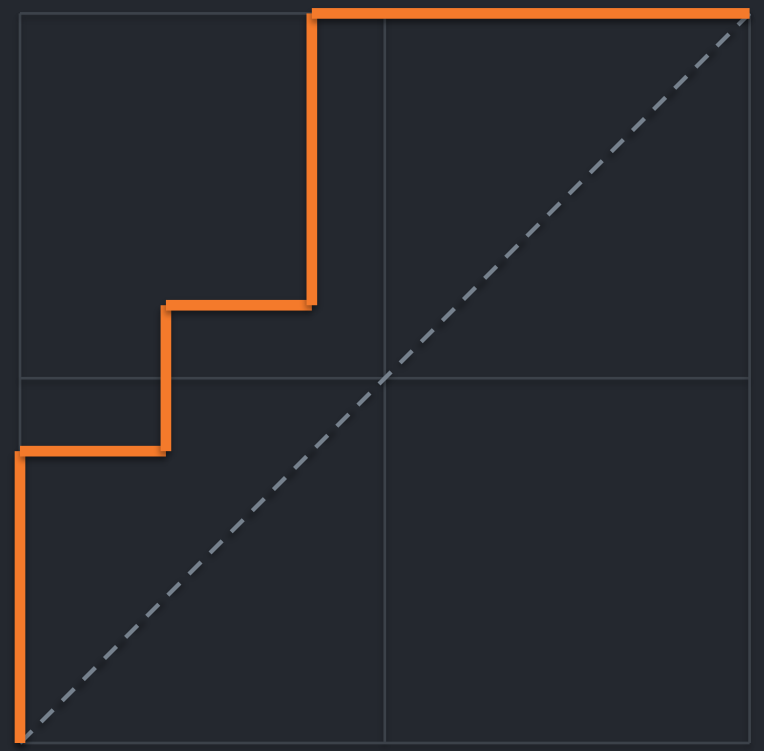
Practical machine learning

# ROC curve and AUC

One example. Ten patients.
Every number explained.



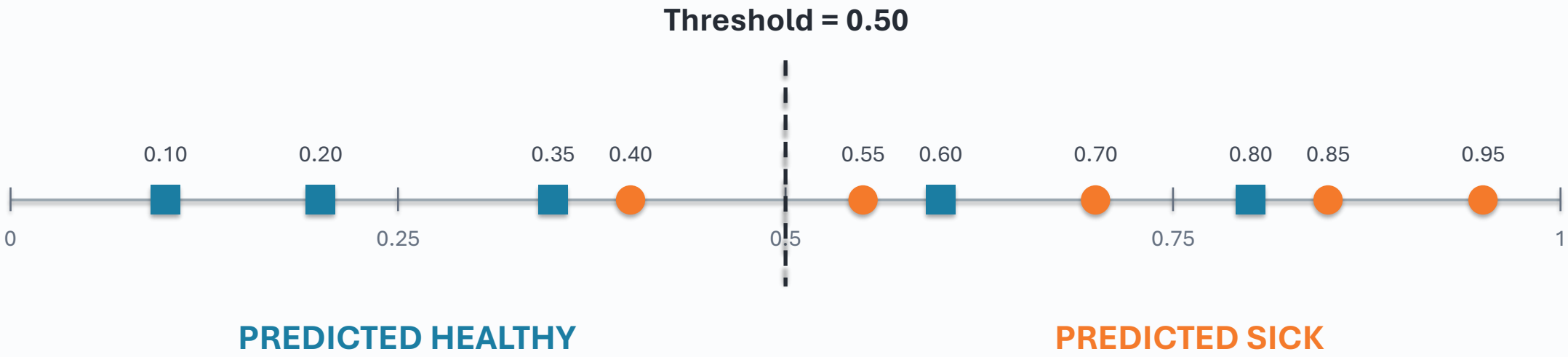
**Actual ROC curve**

Dotted line = random baseline

**AUC = 0.80 · what does it actually mean?**

# A threshold converts scores into predictions

Orange circles = actually sick · Blue squares = actually healthy




## Model score

A larger score means the model considers sickness more likely.

## Decision rule

Score ≥ 0.50 → predict sick. Score < 0.50 → predict healthy.

# At threshold 0.50, count the four outcomes

5 patients actually sick · 5 actually healthy

| | Predicted sick | Predicted healthy | |
|---|---|---|---|
| **Actually sick** | **TP = 4**<br>Caught sick patients | **FN = 1**<br>Missed sick patient | **TPR = 4 / 5 = 0.80**<br>Share of sick patients caught |
| **Actually healthy** | **FP = 2**<br>False alarms | **TN = 3**<br>Correctly healthy | **FPR = 2 / 5 = 0.40**<br>Share of healthy patients flagged |

**One ROC point = (FPR, TPR) = (0.40, 0.80)**

# Different thresholds produce different ROC points

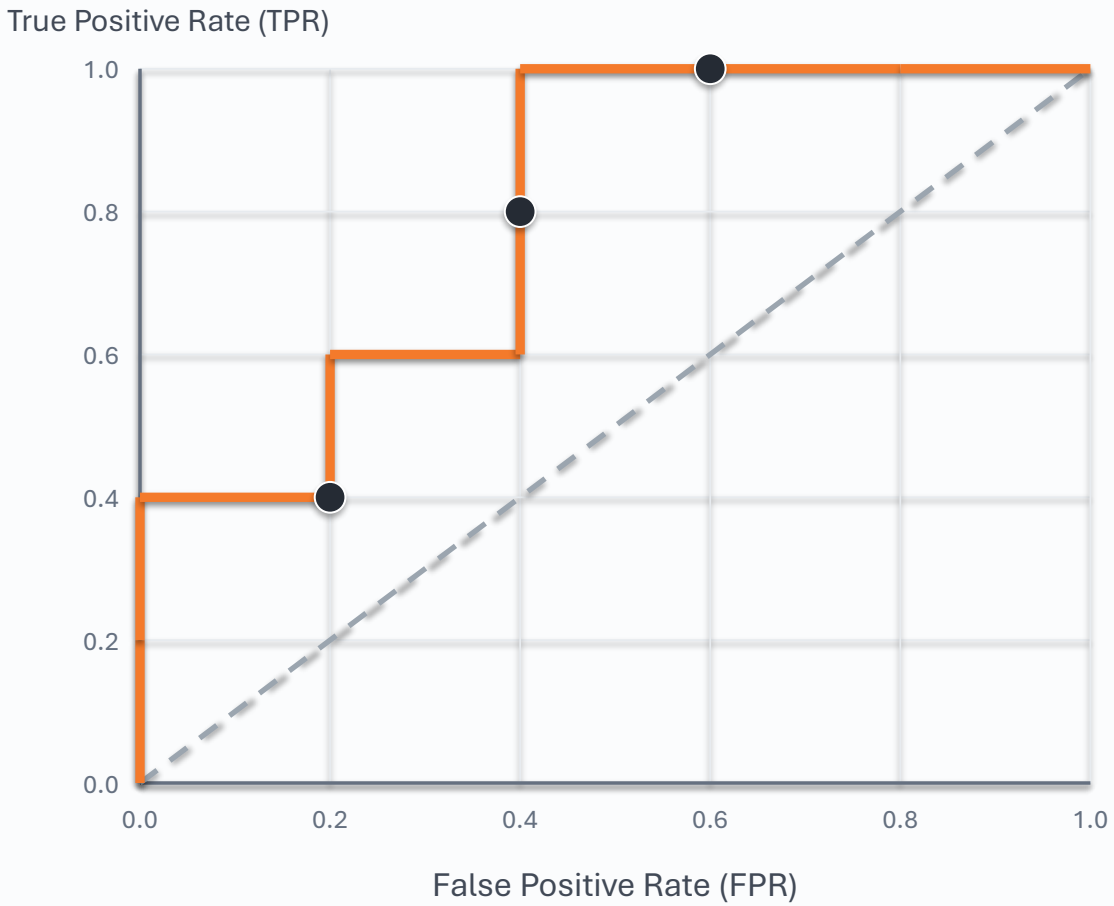
A lower threshold catches more sick patients but can also trigger more false alarms.

| Threshold | Sick caught TP | Healthy flagged FP | TPR | FPR |
|---|---|---|---|---|
| **0.75** | 2 of 5 | 1 of 5 | 0.40 | 0.20 |
| **0.50** | 4 of 5 | 2 of 5 | 0.80 | 0.40 |
| **0.30** | 5 of 5 | 3 of 5 | 1.00 | 0.60 |

**Each threshold gives a coordinate (FPR, TPR): (0.20, 0.40) · (0.40, 0.80) · (0.60, 1.00)**

# Plot FPR on the x-axis and TPR on the y-axis

For 10 patients, the complete curve is a staircase as we change the threshold.




**At threshold 0.75**

(FPR, TPR) = (0.20, 0.40)

**At threshold 0.50**

(FPR, TPR) = (0.40, 0.80)

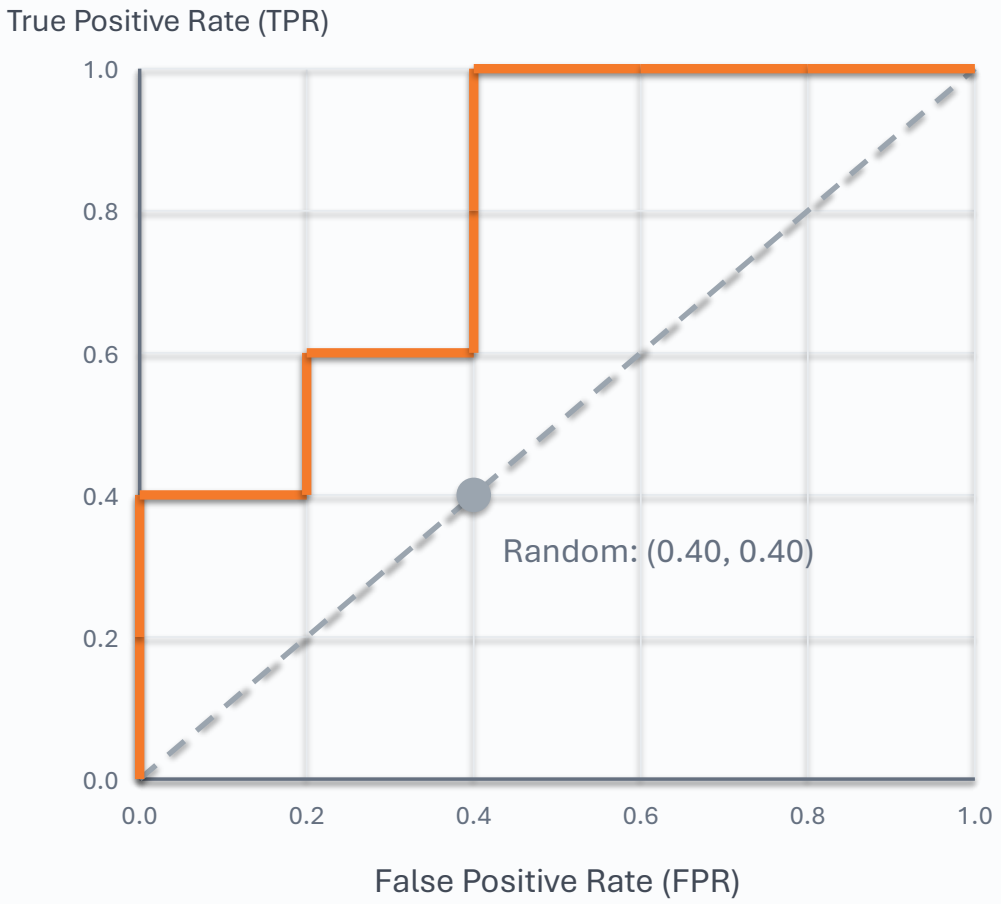
**At threshold 0.30**

(FPR, TPR) = (0.60, 1.00)

The remaining corners come from the other patient scores.

# What does the dotted diagonal mean?

It is the random-guessing reference line - not our model's ROC curve.




**Dotted line: random baseline**

At FPR = 0.40, its TPR is 0.40.

**Orange staircase: our model**

At threshold 0.50: FPR = 0.40, TPR = 0.80.

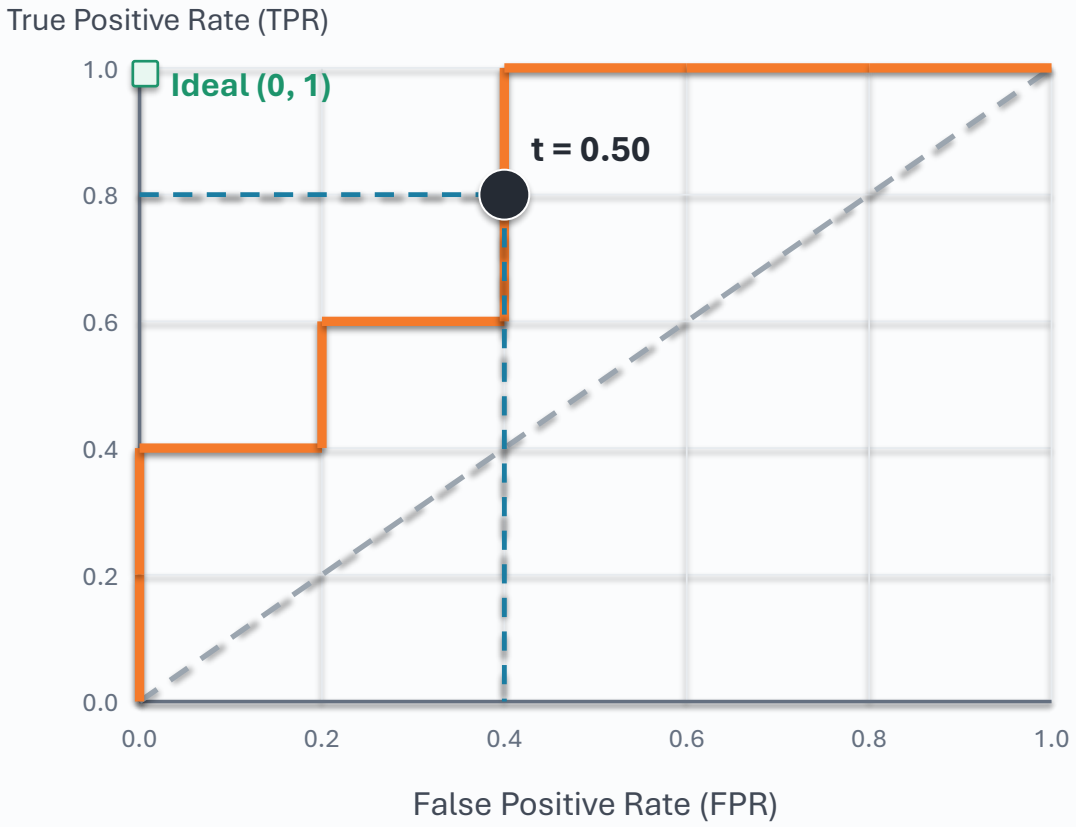
**Ideal ROC point: (0, 1)**

TPR = 1: catch all sick. FPR = 0: no false alarms.

**Random baseline AUC = 0.50**

# How do we read the ROC curve?

Read one point first. Then judge how the whole curve behaves.




### 1 Read the x-axis: FPR = 0.40

2 of 5 healthy patients are wrongly flagged.

### 2 Read the y-axis: TPR = 0.80

4 of 5 sick patients are correctly caught.

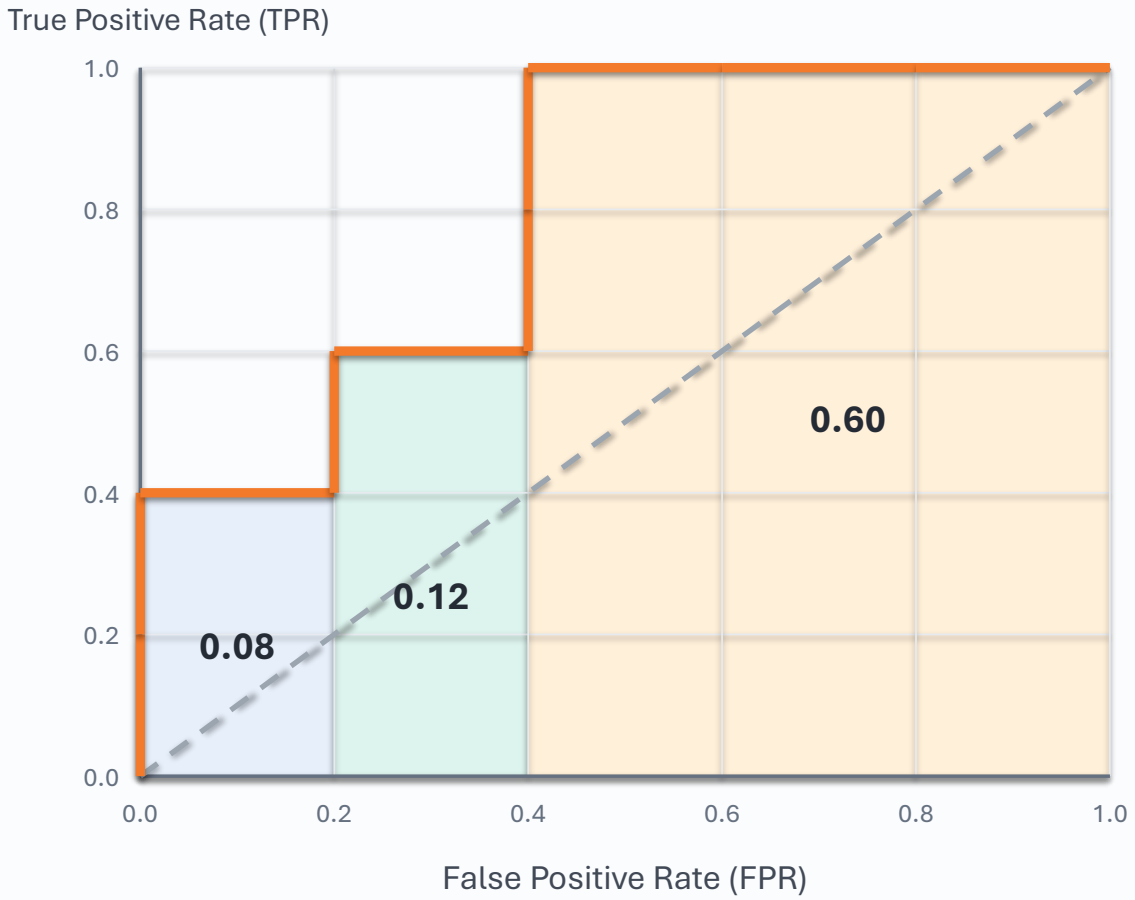
### 3 Judge the curve's position

Near (0, 1) = stronger classification.
Above the diagonal = better than random.

**Top-left is ideal; the best practical threshold also depends on the cost of each mistake.**

# AUC = add the three shaded rectangles

AUC is the area under the full ROC staircase - not just the three example points.




**Rectangle 1**

0.20 × 0.40 = 0.08

**Rectangle 2**

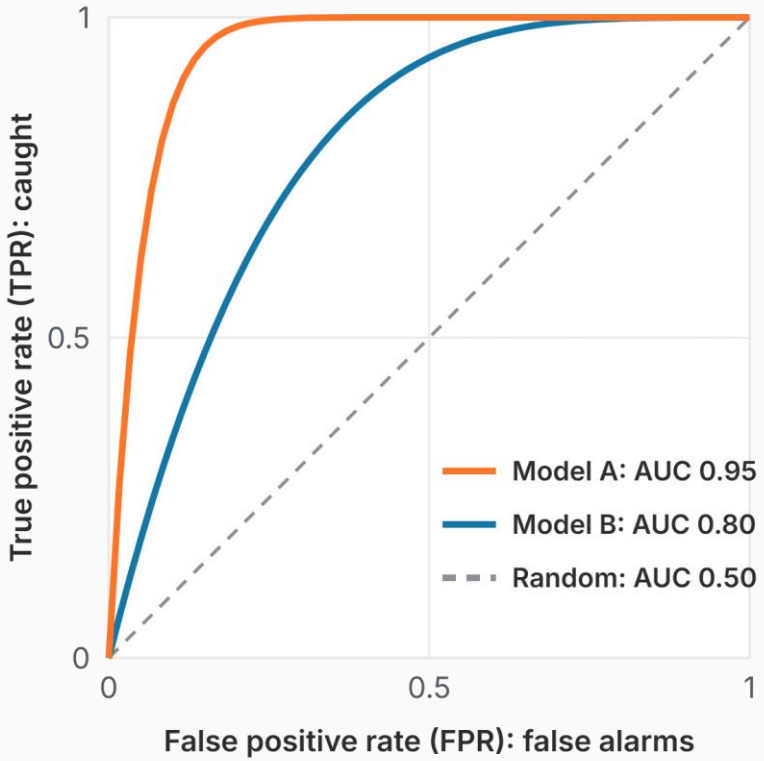
0.20 × 0.60 = 0.12

**Rectangle 3**

0.60 × 1.00 = 0.60

**AUC = 0.08 + 0.12 + 0.60 = 0.80**

# Comparing models with ROC and AUC




### Closer to the top-left wins

Model A's curve is above Model B's everywhere, so A is better at every threshold. Its larger AUC says the same in one number.

### AUC picks the model, not the threshold

After choosing a model, pick the threshold on its curve that fits the cost of each mistake.